

Quantifying Explainability and Applying it to Quanvolutional Neural Network

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Abstract—Neural networks for image classification play an integral role in different domains such as security, healthcare, and medicine. Explaining the decisions of such neural networks is a key challenge due to the complexity of the models, which are often black box in nature. While several metrics for explainability quantification have been previously proposed, there remains a need for an intuitive aggregate measure tailored to image classification tasks. This paper introduces a Frobenius norm-based quantification for explainability and demonstrates its effectiveness on Quanvolutional Neural Networks (QNNs). QNNs are picked because they integrate principles from quantum computing into neural network design and represent a novel class of models whose explainability has not yet been thoroughly explored. As quantum machine learning continues to advance, understanding the behavior of quantum-enhanced models like QNNs is crucial for building trust, enhancing transparency, and guiding future development. Experimental results suggest that the proposed metric offers valuable insights into QNN's behavior, potentially aiding model developers in improving explainability.

I. INTRODUCTION

Quantum computing has shown promising advantages for solving classically intractable problems [1] [2] [3] [4]. The advent of variational quantum algorithms has enabled research demonstrating the capability of hybrid quantum-classical machine learning models [5]. One notable hybrid framework is the Quanvolutional Neural Network (QNN), which differs from traditional Convolutional Neural Networks (CNNs) by replacing the classical, linear convolutional filters with quantum circuits. These quantum circuits encode image patches into a high-dimensional space, utilizing quantum phenomena such as superposition and entanglement to extract features in a potentially more efficient and expressive manner [6].

The kernels in a Convolutional Neural Network (CNN) are linear as they perform a linear operation for feature extraction when applied to the input data. A quantum kernel may extract the same features in fewer operations or in a more compact form, as quantum states can encode classical data in higher-dimensional space, leading to fewer operations than what a classical CNN might require for the same task [6]. While the potential for QNNs to outperform classical largely depends on the advancement of quantum hardware, it is important to study their behavior and explainability to better understand their strengths and limitations. In classical deep learning, explainability has become essential not only for building trust but also for debugging, improving models, and ensuring reliability in sensitive applications [7]. As quantum

machine learning evolves, similar needs arise for quantum-enhanced models.

This work explores the explainability of Quanvolutional Neural Networks (QNNs) by proposing a novel mathematical approach for quantifying explainability. The rest of the paper is organized as follows: Section II provides an overview of related works for QNN and the explainability. Section III introduces concepts in quantum computing and QNNs. Section IV describes the proposed quantification of explainability. Section V describes the setup of QNNs and datasets. The results of the experiments are presented and discussed in section VI. Finally, the concluding discussions and remarks about future works are in Section VII.

II. RELATED WORKS

A. Quantifying Explainability

There have been several prior works that proposed quantifying explainability for different domains and different models. A roadmap on how to quantify explainability was laid out in [8], in which the authors propose a framework by assigning scores to three core metrics of an explanation: faithfulness: how well an explanation reflects model behavior; stability: how consistent explanations are for similar inputs; and comprehensibility: how interpretable the explanations are to humans. Using these metrics, they evaluate popular methods like LIME [9] and SHAP [10] with various supervised ML models for mortgage bankruptcy prediction. Their results show that no single method scores best across all metrics, highlighting inherent trade-offs in explainability.

The authors of the paper [11] proposed a new metric for explainability called Degree of Explainability (DoX) through pre-trained deep language models for general-purpose answer retrieval for existing XAI (Explainable AI models). Their DoX could objectively quantify how much of the information is explained by an XAI by measuring its relevance to answering a pre-defined set of archetypal questions. Their results in heart disease prediction and credit approval decision-making using regression and ANN (Artificial Neural Networks) showed that the DoX was able to quantify explainability in natural language information corpora. However, their method cannot be used directly on a black-box model to understand how much of it is explainable and only works with the output of an ensemble of XAI algorithms.

Various methods of generating explanations accessible to pathologists for breast cancer sub-typing using GNNs (Graph Neural Networks) were proposed in [12]. Additionally, the authors also proposed a set of novel quantitative metrics based on statistics of class separability using pathologically measurable concepts to characterize explanations. They observed that the explanation quality is directly proportional to the GNN's classification performance for a pair of classes. Their proposed metrics are also applicable in other domains by using domain-specific intuitive concepts.

A training strategy to enforce a periodic feature-importance-based feedback to encourage the deep neural networks to focus on the image regions that directly correspond to the ground-truth object for object detection was proposed in [13]. Their approach leads to more explainable classification models and better generalization to unseen domains. Additionally, it had no association with accuracy degradation on the original domain and no added test-time complexity.

B. Hybrid Quantum-Classical Neural Networks

Quantum Convolutional Neural Networks proposed in [6] used quantum circuits as a kernel for feature extraction, i.e., a quantization, the quantum analog of convolution in CNNs. Their results on MNIST showed that the QNN models had both higher test set accuracy and faster training compared to the purely classical CNNs. Variational Quantum Convolutional Neural Networks (VQNNs) proposed in [14] modify QNNs by adding trainable variational quantum circuits to the network architecture, which significantly improved the VQNNs' performance compared to QNNs on MNIST dataset.

AdvQNN proposed in [15] evaluated the robustness of QNNs against the following attacks: Fast Gradient Sign Method (FGSM), Projected Gradient Descent (PGD), and Momentum-based Iterative Method (MIM). Their evaluation on QNNs achieved up to 60% higher robustness for the MNIST and 40% for FMNIST datasets compared to CNNs.

A CNN-based custom RANet model to detect rheumatoid arthritis (RA) using thermal images, which show temperature differences in the hand to identify abnormalities associated with RA, was proposed in [16]. The authors compared the performance of the RANet model with a QNN for the same task. The RANet and QNN architectures achieved the highest accuracy levels, which were 95% and 93.33%, respectively, demonstrating that the RANet model and QNN could be employed as a pre-screening diagnostic tool to evaluate RA.

By comparing previous related works, the major innovations of this work are concluded as follows:

- Proposing an intuitive quantification of explainability for neural networks for image classification tasks applicable to any domain.
- Applying the proposed metric on QNN and demonstrating QNN's explainability for the first time.

III. BACKGROUND

A. Quantum Computing

The quantum analog of a classical bit is a qubit, denoted by Dirac notation [17] as follows:

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

Unlike a classical bit, a qubit can exist in both $|0\rangle$ and $|1\rangle$ simultaneously, called a superposition state:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

where α, β are complex numbers called probability amplitudes obeying $|\alpha|^2 + |\beta|^2 = 1$. However, a qubit can only exist in a superposition of $|0\rangle$ and $|1\rangle$ before measurement. Following the measurement, it will collapse to the state $|0\rangle$ or $|1\rangle$ with probability $|\alpha|^2$ or $|\beta|^2$, respectively. Measurement is a process of extracting information from the quantum state, e.g., an electron can exist in multiple places at the time, which can be described by a quantum state, and following the measurement, the electron will be found in one region or another when an experiment is performed to locate it.

A qubit can be represented by a wire (single horizontal line) in a quantum circuit and can be measured to a classical bit, also represented by a wire (double horizontal line). The measurement is shown as a meter symbol inside a box. A qubit being measured is shown in Fig. 1.

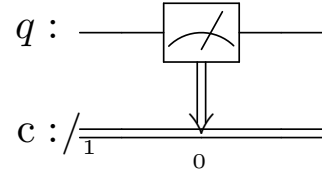


Fig. 1: Quantum circuit for a single qubit q being measured to a classical bit c

Qubits can be visualized as a point on a 3-dimensional *Bloch* sphere. Because $|\alpha|^2 + |\beta|^2 = 1$, state $|\psi\rangle$ can be rewritten as:

$$|\psi\rangle = \cos\frac{\theta}{2}|0\rangle + e^{i\phi}\sin\frac{\theta}{2}|1\rangle$$

where θ and ϕ are polar and azimuthal angles of $|\psi\rangle$ on the *Bloch* sphere, respectively, as shown in Fig. 2.

The state of a qubit can be manipulated using single-qubit gates, which are 2×2 matrices. After applying a single qubit gate, the final state is obtained by multiplying the gate matrix by the initial state. Table I shows examples of single-qubit gates that transform a qubit state by rotating it.

The other unique property of a quantum state is the entanglement; an intuitive example is the Bell state $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$

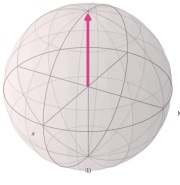
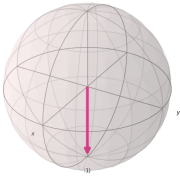
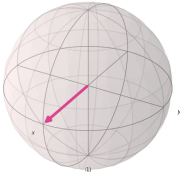
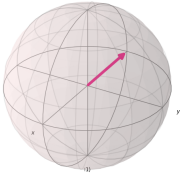
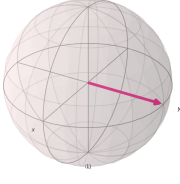
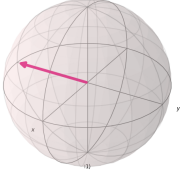
Single-Qubit Gate	Circuit & Description	Initial State	Transformation	State After
$R_x(\theta) = \begin{pmatrix} \cos \frac{\theta}{2} & -i \sin \frac{\theta}{2} \\ -i \sin \frac{\theta}{2} & \cos \frac{\theta}{2} \end{pmatrix}$	$q : \text{---} \boxed{R_X(\theta)} \text{---}$ R_X rotates a qubit around X-axis by θ	 $ 0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$	$R_x(\pi) 0\rangle = -i 1\rangle \rightarrow$ $\begin{pmatrix} 0 & -i \\ -i & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = -i \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ $-i$ is a global phase and can be ignored	 $ 1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$
$R_y(\theta) = \begin{pmatrix} \cos \frac{\theta}{2} & -\sin \frac{\theta}{2} \\ \sin \frac{\theta}{2} & \cos \frac{\theta}{2} \end{pmatrix}$	$q : \text{---} \boxed{R_Y(\theta)} \text{---}$ R_Y rotates a qubit around Y-axis by θ	 $ +\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$	$R_y(\pi) +\rangle = -\rangle \rightarrow$ $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix} = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{pmatrix}$	 $ -\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$
$R_z(\theta) = \begin{pmatrix} e^{-i\frac{\theta}{2}} & 0 \\ 0 & e^{i\frac{\theta}{2}} \end{pmatrix}$	$q : \text{---} \boxed{R_Z(\theta)} \text{---}$ R_Z rotates a qubit around Z-axis by θ	 $ i\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix}$	$R_z(\pi) i\rangle = -i -i\rangle \rightarrow$ $\begin{pmatrix} -i & 0 \\ 0 & i \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{i}{\sqrt{2}} \end{pmatrix} = -i \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{-i}{\sqrt{2}} \end{pmatrix}$ $-i$ is a global phase and can be ignored	 $ -i\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -i \end{pmatrix}$

TABLE I: Examples of single qubit gates rotating a qubit around the *Bloch* sphere.

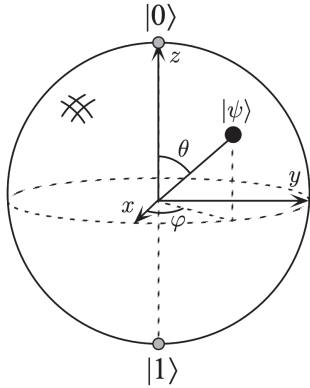


Fig. 2: *Bloch* sphere representation of a qubit [17]

where the state of one qubit depends on the state of the other, e.g., if the first qubit is measured in state $|0\rangle$, the state of the second qubit changes to $|0\rangle$ without even measuring it. Entanglements can be generated by using two-qubit gates like:

$$CNOT = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

Fig. 3 shows a quantum circuit to generate an entangled Bell state.

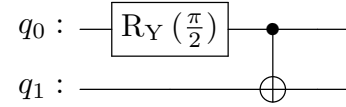


Fig. 3: Quantum circuit using CNOT gate to generate entangled bell state $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$

The two-qubit gates, together with single-qubit gates, provide all of the necessary operations for universal quantum computation [18]. A quantum gate U must be a unitary operation ($UU^\dagger = U^\dagger U = I$).

B. Quanvolutional Neural Network

A quanvolutional filter of size $n \times n$ performs the following three steps:

- 1) **Encoding:** Each pixel of an image patch of size $n \times n$ is encoded into a quantum circuit with n^2 qubits. QNN implementation in this paper considers two different encoding methods, discussed later in Section V-B.
- 2) **Circuit:** Further computations are performed on the encoded qubit states by using different quantum circuits. In this work, two different entangling circuits are considered, discussed later in Section V-B.
- 3) **Decoding:** All n^2 qubits are measured resulting in n^2 quanvolutional output channels.

IV. QUANTIFYING EXPLAINABILITY

A heatmap is a 2D matrix showing how important each image region, i.e., features are in neural network's making a decision. Higher values (brighter pixels) mean the feature played high importance in the model's decision. Grad-CAM (Gradient-weighted Class Activation [19]) is used to generate heatmaps for QNNs in this paper.

The explainability $\mathcal{E}_{NN}$ of a neural network is quantified using the Frobenius norm. Frobenius norm of a $m \times n$ matrix A is:

$$\|A\|_F = \sqrt{\sum_{i=1}^m \sum_{j=1}^n |A_{ij}|^2}$$

$\mathcal{E}_{NN}$ captures the similarity between the difference of the heatmap of a neural network and an ideal heatmap:

$$\mathcal{E}_{NN} = \|L_{Ideal} - L_{NN}\|_F$$

where $\|\cdot\|_F$ represents the Frobenius norm, L_{NN} is a heatmap of a neural network generated using Grad-CAM [19] algorithm, and L_{Ideal} is an ideal heatmap showing what features should be used to make classifications, generated using existing highly explainable neural networks [20].

V. EXPERIMENTAL SETUP

A. Dataset

Tiny-ImageNet dataset is selected, given the goal of analyzing explainability for real-world situations.

- Tiny-ImageNet dataset downloaded from Kaggle [21] with 200 classes, 500 64x64 RGB images in each class
- Four classes are chosen for the purpose of demonstrating the explainability of QNN, as shown in Fig. 4, along with the reasoning for choosing said classes.

B. QNN-Structure

Fig. 5 shows the architecture of QNN with input 3-channel RGB image passing through a 2x2 quanvolutional filter under different encoding and entanglement configuration settings to produce a 4-channel output, followed by a fully-connected trainable dense layer.

Two different encoding schemes are considered, as shown in Fig. 6 and described below:

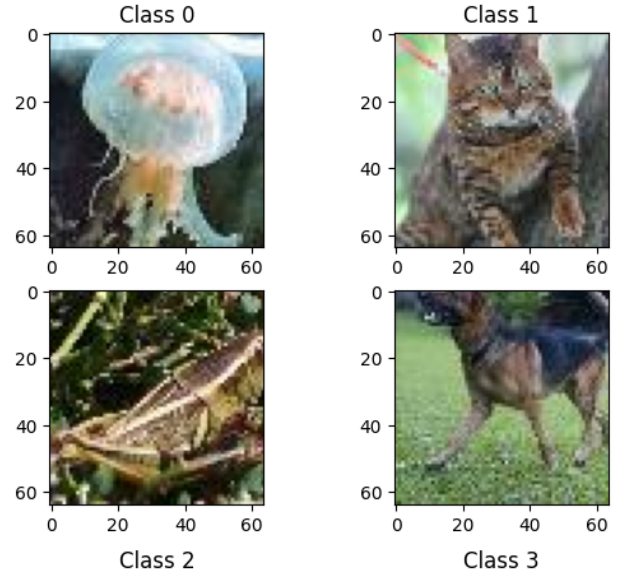


Fig. 4: Four Tiny-ImageNet classes (Jellyfish, Cat, Grasshopper, and Dog) chosen to demonstrate explainability $\mathcal{E}_{QNN}$ for QNNs. This choice is deliberate since the Jellyfish class contains images with mostly blue/black backgrounds, thus, a model might become biased, attaining higher $\mathcal{E}_{QNN}$.

- 1) **Angle:** rotate the qubit where the rotation angle is the average of RGB pixel values shown in Fig. 6a.
- 2) **Amplitude:** use (R, G, B) pixel values as (x, y, z) coordinate of the mapped qubit shown in 6b. The mapped state on the *Bloch* sphere is shown in Fig. 6c.

As discussed earlier, entangling quantum circuits are applied after encoding. Stronger entanglement allows the quantum system to capture deeper and more intricate dependencies that classical models might miss [22]. Two different entanglements considered are:

- 1) **Basic:** cheaper as it has less number of *CNOT* but less entanglement between different pixels shown in Fig. 7a
- 2) **Strong:** expensive but higher entanglement between different pixels, hence, theoretically, generates better feature maps shown in Fig. 7b

Encoding	Entanglement	Optimizer	Best-Score
Angle	Basic	NADAM	0.4476
Angle	Strong	ADAM	0.471
Amplitude	Basic	NADAM	0.542
Amplitude	Strong	NADAM	0.543

TABLE II: Results from GridSearchCV (2-fold cross-validation) with different optimizers for different encoding and entanglement configuration

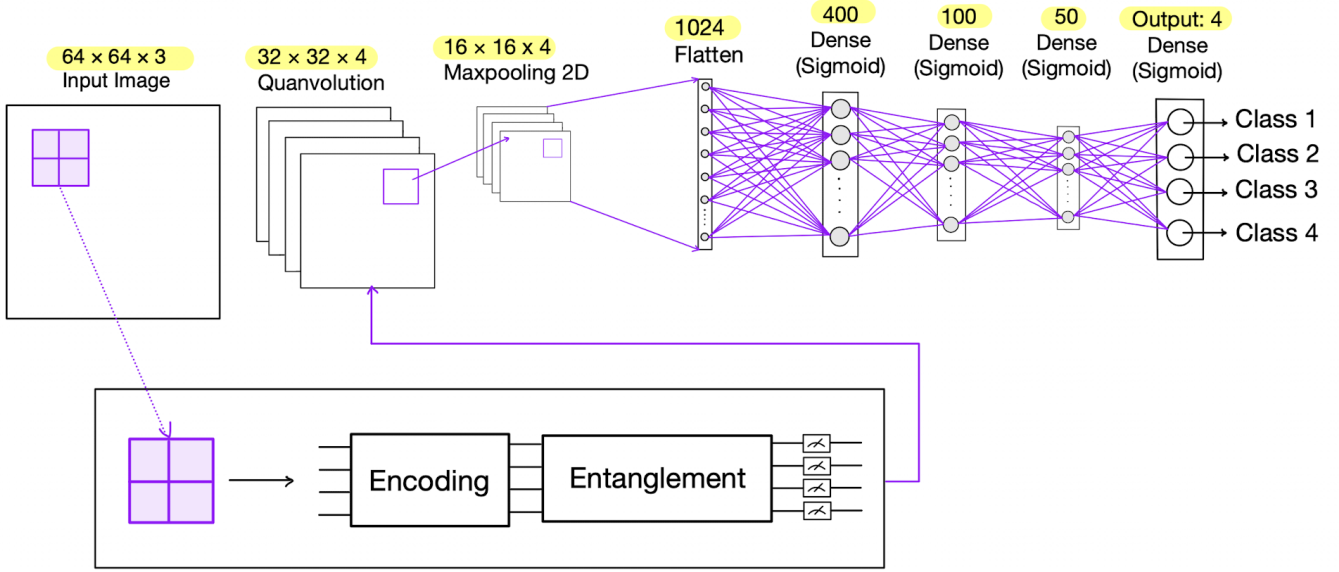


Fig. 5: Flow of an input image undergoing 2×2 quanvolutional filter followed by a fully-connected layer.

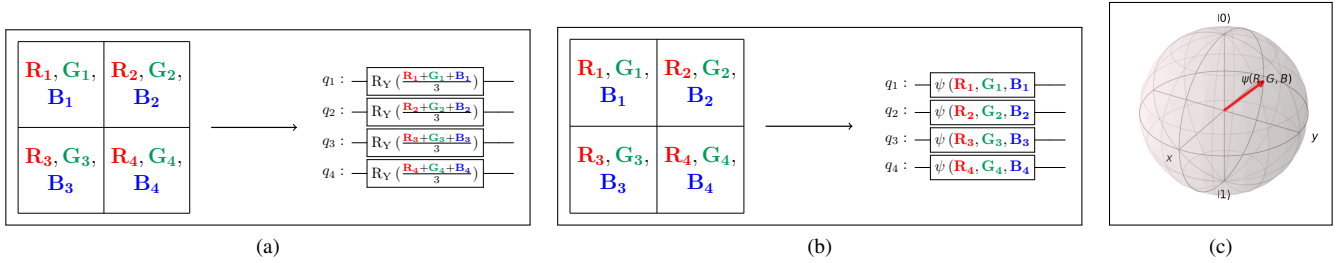
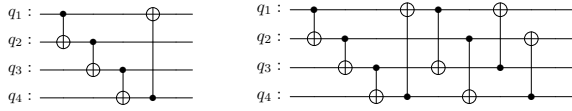


Fig. 6: (a) **Angle encoding**: rotate qubits using mean of pixel values. (b) **Amplitude encoding**: use normalized RGB pixel values to map qubits on the *Bloch* sphere. (c) Amplitude encoding shown on the *Bloch* sphere



(a) Basic Entanglement

(b) Strong Entanglement

Fig. 7: Two different entanglement circuits applied after encoding

C. QNN-Training

GridSearchCV is used to find the optimal optimizer from (adam,nadam,sgd) with two-fold cross-validation. Results from the GridSearchCV are shown in Table II.

Given the extremely slow nature of GridSearchCV, manual experiments are performed with different activation functions (sigmoid, softmax, tanh, relu), and sigmoid is found to be better compared to others.

Ten different QNN models under the four configurations,

as mentioned in Table II, are trained for 500 iterations with batch size 32 and 75% random train-test-split.. Fig. 8 shows the plot of accuracy and loss averaged for training and validation of ten QNNs with amplitude encoding and basic entanglement.

All models are trained on a Macbook Air laptop with 8GB RAM and an M2 processor. The code for all the experiments performed in this paper is available on GitHub: explaining_quanvolution.

D. Ideal Heatmap Generation

Explainable CNN proposed in [23] employed encoder-decoder neural networks in a CNN architecture to represent regions of interest in an image based on its category. They performed experiments on CIFAR, MNIST, and Tiny-ImageNet and also made their pre-trained models available on GitHub, which are used for calculating $\mathcal{E}_{QNN}$ on the Tiny-ImageNet dataset.

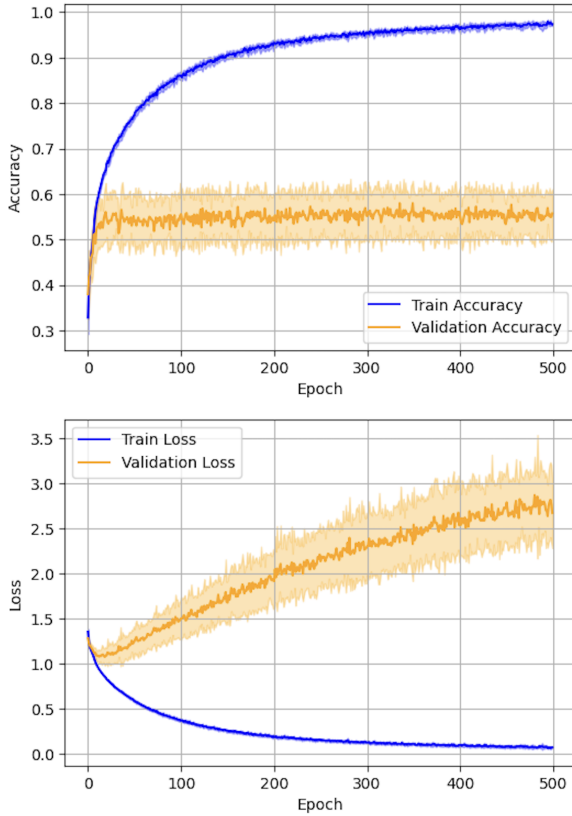


Fig. 8: Accuracy and Loss averaged for training and validation of ten QNNs with amplitude encoding and basic entanglement. The plateauing of validation accuracy and increasing validation loss suggests QNN is overfitting.

VI. RESULTS & DISCUSSION

Table III shows the comparison of accuracy, F1-score, and $\mathcal{E}_{QNN}$ under different configurations, which do not seem to have a significant impact on QNN’s performance and explainability. Henceforth, results will be discussed for the amplitude encoding and basic entanglement configuration. The results of the remaining configurations follow similar trends and are presented in Appendix VII.

From Fig. 9, the explainability of all QNN models is significantly worse for the Jellyfish class, which suggests that the QNN is relying on blue/black backgrounds to make a classification, as illustrated in Fig. 10. Thus, the proposed quantification of explainability $\mathcal{E}_{NN}$ is able to provide insights into how model’s performance in terms of accuracy and explainability changes for individual classes. High accuracy and high F1-score do not always equate to high explainability.

VII. CONCLUSION

This paper demonstrated how a novel Frobenius norm-based quantification of explainability can provide insights into aggregate as well as class-wise explainability of a model. The proposed metric is intuitive and can be applied to any other

Encoding	Entanglement	Accuracy	F1-Score	$\mathcal{E}_{QNN}$
Angle	Basic	0.8758 $\pm$ 0.004	0.8756 $\pm$ 0.003	22.184 $\pm$ 0.108
Angle	Strong	0.8741 $\pm$ 0.003	0.874 $\pm$ 0.003	22.28 $\pm$ 0.101
Amplitude	Basic	0.8806 $\pm$ 0.004	0.8807 $\pm$ 0.004	23.837 $\pm$ 0.184
Amplitude	Strong	0.8772 $\pm$ 0.005	0.8775 $\pm$ 0.005	23.444 $\pm$ 0.208

TABLE III: Accuracy, F1-Score and $\mathcal{E}_{QNN}$ of QNNs under different configurations

neural network for image classification tasks. However, our quantification of explainability further needs to be investigated on its robustness, stability, comprehensibility, and feasibility, as discussed in [8]. We plan to evaluate more neural networks using our proposed metric of explainability.

Experiments with a larger QNN filter (5x5) with more complex datasets like CIFAR, ImageNET, etc., investigate why QNN might be overfitting as seen in Fig. 8, and how to make QNNs more explainable, are planned for future works. Additionally, “Quantum deep convolutional neural networks [24], [25]” were proposed as a complete quantum analog for CNNs, and existing explainability techniques cannot be applied to these quantum convolutional neural networks. We plan to research how to explain them and quantify their explainability in the future.

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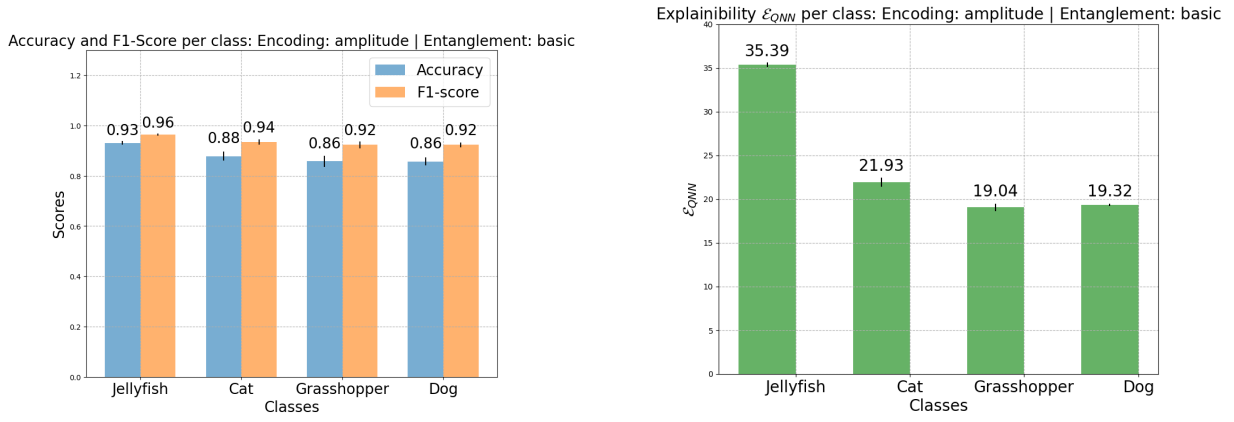


Fig. 9: Class-wise comparison of Accuracy, F1-Score and $\mathcal{E}_{QNN}$ for QNN models amplitude encoding and basic entanglement configuration. The explainability plot on the right suggests that the QNN model is biased and relies on blue/black backgrounds in jellyfish images, leading to higher $\mathcal{E}_{QNN}$.

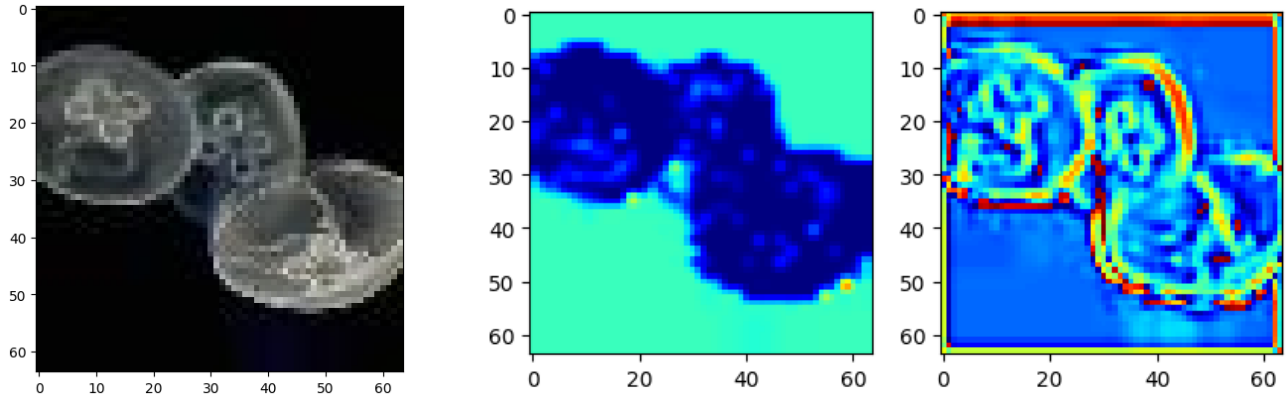
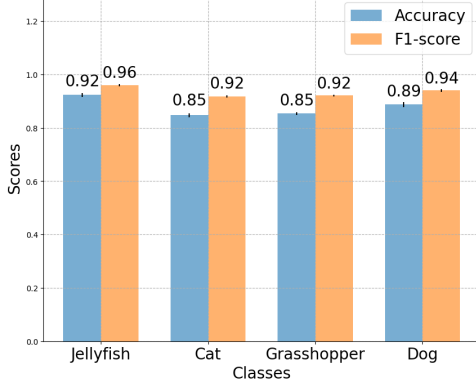


Fig. 10: **Left:** is the original Jellyfish image. **Middle:** image is the Grad-CAM heatmap for QNN, which relies on the black background to make the classification. **Right:** image is a heatmap for encoder-decoder-based CNN [23] which uses actual jellyfish features for classifications.

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APPENDIX

Accuracy and F1-Score per class: Encoding: angle | Entanglement: basic



Explainability $\mathcal{E}_{QNN}$ per class: Encoding: angle | Entanglement: basic

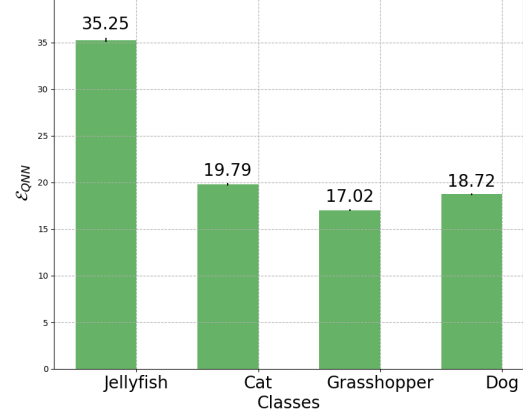
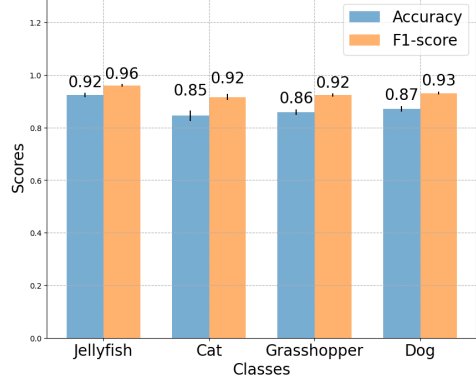


Fig. 11: Class-wise comparison of Accuracy, F1-Score and $\mathcal{E}_{QNN}$ for QNN models angle encoding and basic entanglement configuration

Accuracy and F1-Score per class: Encoding: angle | Entanglement: strong



Explainability $\mathcal{E}_{QNN}$ per class: Encoding: angle | Entanglement: strong

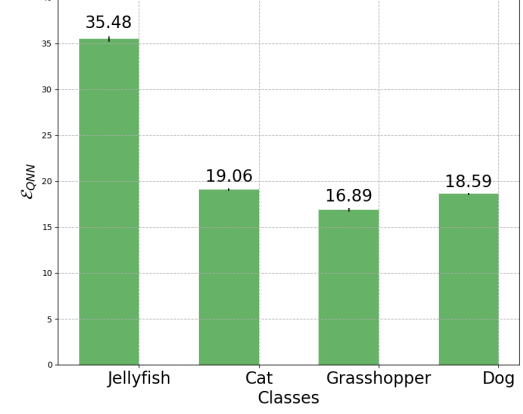
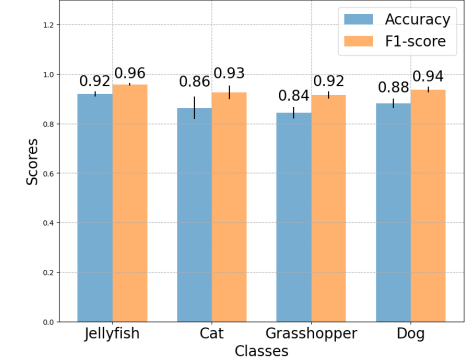


Fig. 12: Class-wise comparison of Accuracy, F1-Score and $\mathcal{E}_{QNN}$ for QNN models angle encoding and strong entanglement configuration

Accuracy and F1-Score per class: Encoding: amplitude | Entanglement: strong



Explainability $\mathcal{E}_{QNN}$ per class: Encoding: amplitude | Entanglement: strong

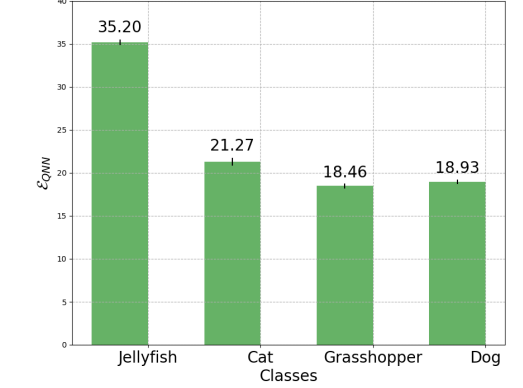


Fig. 13: Class-wise comparison of Accuracy, F1-Score and $\mathcal{E}_{QNN}$ for QNN models amplitude encoding and strong entanglement configuration